

① Inputs

Natural language parsing

e.g. Expressway, heavy traffic, cold region, clay subgrade, budget limited...

LLM parsing
structured extraction

or

Structured design input

- Design Requirements
- Service life
- Load conditions
- Climate & environment
- Materials & availability
- Construction constraints
- Budget envelope
- ...



② Agent optimization loop

2.1 Generator (LLM)

- Interpret requirements
- Retrieve relevant clauses
- Propose qualitative guidance
- No numerical magnitudes

RAG grounding

- retrieval → reranking/fusion
- Multi-region specification clauses
 - Typical structures
 - Lessons & prior designs

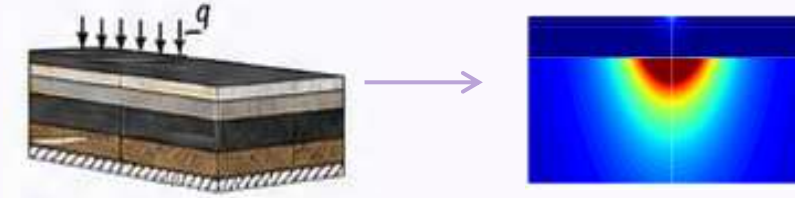
Updated structure

- Updated layer thickness h_i
- Updated layer moduli E_i

Initial state S_0

2.2 Physical verification

Finite element modeling

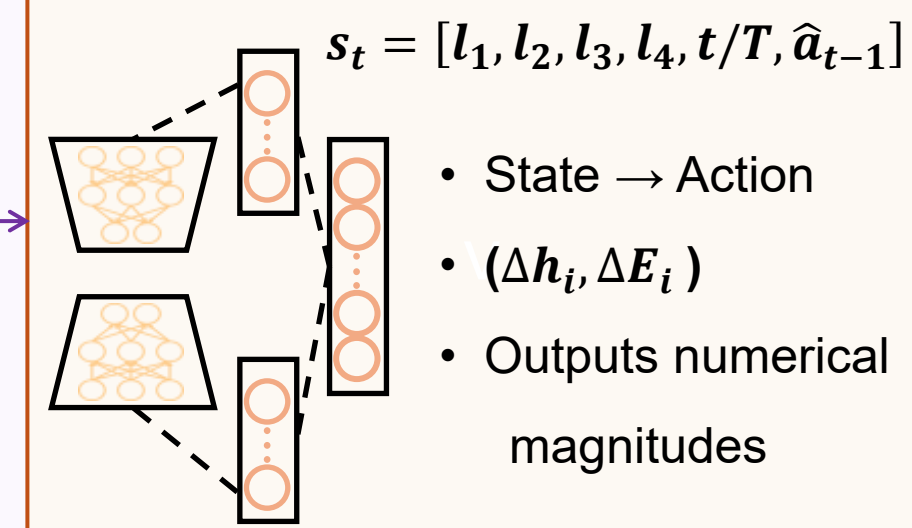


Structural responses

- Stresses, strains, deflections
- Safety factors / displacements
- Other domain-specific responses

PPO state

2.3 PPO Policy Update



PPO's action
 a_{PPO}

LLM's suggestion a_{LLM}

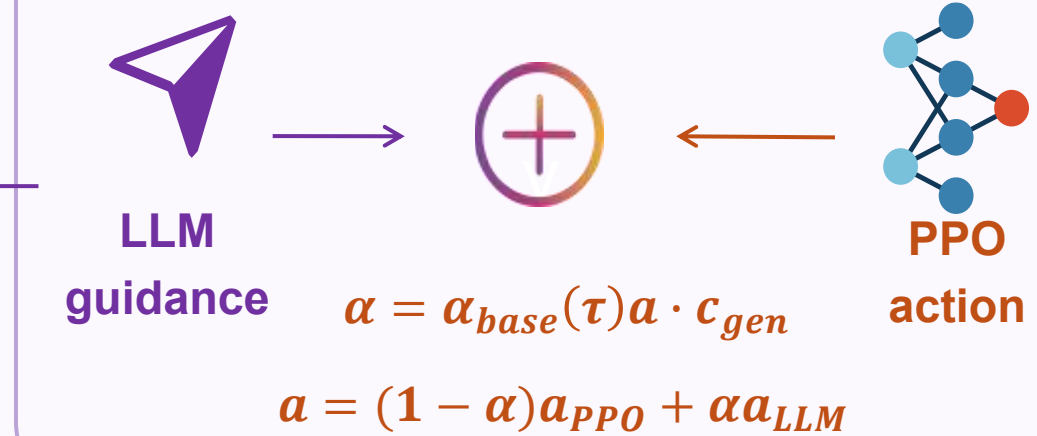
2.5 Numerical guard

Check bounds & construction



- Pass → accept
- Fail → reject / clip

2.4 Action fusion and state update

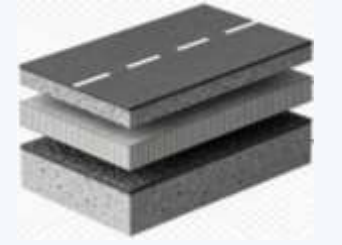


Next state S_{t+1}

③ Outputs

Design Output

- Final structure ($h_1 - h_5, E_1 - E_5$)
- DSR / SCR and B1-B4 margins
- Construction cost and 20 year LCC
- FEA report
- Human review / approve record



Audit Chain

state, action, response, reward, hash (append-only)



④ Audit Chain

Record t-1

state, action, response, reward, hash

Record t

state, action, response, reward, hash

Record t+1

state, action, response, reward, hash

...

state, action, response, reward, hash

⑤ Civil structure candidates



Pavement

validated in this study

- Layer thicknesses & moduli
- B1-B4 safety margins
- Cost, LCC & audit trail



Slope

Not instantiated

- Domain-specific inputs
- Physics-based stability
- Candidate output only



Building

Not instantiated

- Beam-column layout
- Stress / drift / serviceability
- Code checks & audit trail



Foundation

Not instantiated

- Pile / footing parameters
- Bearing & settlement
- Code checks & audit trail